The Three dS_v-Block Decomposition of $\mathbb{R}^9$

Discharging Obligation O-3Gen-Block (NFC-3)

A Suite-Internal Combinatorial Proof

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Discharge record. This paper discharges obligation O-3Gen-Block (Paper D2, §5.3; ledger entry NFC-3). It provides a suite-internal proof that the pendant operator algebra $\mathfrak{g} \cong \mathfrak{gl}(3, \mathbb{R})$ acts block-diagonally on $\mathbb{R}^9 = B_{+1} \oplus B_{-1} \oplus B_0$, where the three blocks are the dS_v-eigenspaces at $K_0 = 7$, and the action on each block is the same $\mathfrak{gl}(3, \mathbb{R})$ matrix. This is conditional on the C-PA-Chev catalog entry (Paper D2) which identifies the pendant algebra as $\mathfrak{gl}(3, \mathbb{R})$. Upon acceptance, entry BM1 of Paper I R4 loses its NFC† flag.

Abstract

We prove that the NF2 class space at $K_0 = 7$ decomposes as $\mathbb{R}^9 = B_{+1} \oplus B_{-1} \oplus B_0$ where $B_{dS_v} = \mathrm{span}\{e_k : dS_v(C_k) = dS_v\}$ for $dS_v \in \{+1, -1, 0\}$, each block has dimension 3, and the pendant operator algebra $\mathfrak{g}$ acts on each block by the same matrix. The proof uses only the definitions of the dS_v coordinate, the pendant operators, and the site-type structure of the NFC kernel (Paper A).

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1 Imports and setup

We work with the finite relational framework of Paper A and the pendant algebra identification from the suite.

- **A-Sig** (Paper A, §3): a finite relational substrate $(U, \mathcal{E})$ has sites of two types: u-sites (those carrying the update observable) and v-sites (those carrying the value observable). The NFC substrate at $K_0 = 7$ has both types.
- **C-PA-Chev** (Paper D2 catalog entry, pending hash): The pendant algebra $\mathfrak{g}_{\text{pend}}$ at $K_0 = 7$ satisfies the 18 Chevalley–Serre relations of $\mathfrak{su}(3)$ (residuals $< 10^{-15}$) and is isomorphic to $\mathfrak{gl}(3, \mathbb{R})$ (real form) acting on $\mathbb{R}^9$. Status: Computational for canonical $K_0 = 7$ substrate.
- **BO1** (Paper BO, Theorem 2.1): WL-1 stabilization at $K_0 = 7$ produces exactly $K_0 = 7$ color classes with canonical coloring. Status: Unconditional.
- **BJ-ACE** (Paper BJ): Avoidance-Cycle Exclusion. Under Phase-A, the canonical Phase-A recurrence certificate $T_{\text{cov}}(U) < \infty$. Status: Unconditional.

2 The dS_v coordinate and the nine NF2 classes

Definition 2.1 (dS_v coordinate at $K_0 = 7$)**.** For a NF2 class C_k (a stabilized WL color class at $K_0 = 7$), the *site-value change coordinate* $dS_v(C_k)$ is defined as:

$$dS_v(C_k) \;:=\; \operatorname{sign}\big(\bar{s}_v(C_k^+) - \bar{s}_v(C_k^-)\big) \;\in\; \{+1, -1, 0\},$$

where $\bar{s}_v(C_k^\pm)$ is the mean value of the v-site observable s_v on the successor $(+)$ and predecessor $(-)$ class distributions associated with C_k under admissible extension.

The value $+1$ indicates the v-site observable increases on average under canonical extension; -1 indicates decrease; 0 indicates no change.

Proposition 2.2 (dS_v is well-defined and takes three values)**.** *Under WL-1 stabilization (BO1), the coordinate $dS_v(C_k)$ is well-defined for each of the $K_0 = 7$ color classes. At $K_0 = 7$, each value in $\{+1, -1, 0\}$ is taken by exactly 3 classes. Therefore:*

$$\mathbb{R}^9 \;=\; B_{+1} \oplus B_{-1} \oplus B_0, \qquad \dim B_{dS_v} = 3 \textit{ for each } dS_v.$$

Proof. Well-definedness: By BO1 (WL-1 stabilization), each color class C_k has a canonical coloring $\sigma : \{0, \ldots, K_0 - 1\} \to \{0, \ldots, K_0 - 1\}$ with zero residuals. The mean v-site value $\bar{s}_v(C_k^\pm)$ is therefore a well-defined real number attached to each class, and $dS_v(C_k)$ is its sign.

Three values at $K_0 = 7$: The WL-1 stabilization at $K_0 = 7$ produces the canonical coloring $\sigma : 0, 0, 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1, 0, 0$ (Paper BO, Theorem 2.1 with residuals zero). The site structure at $K_0 = 7$ has a u-v symmetry: for each class C_k with $dS_v = +1$, there is a corresponding class $C_{k'}$ with $dS_v = -1$ (by the u-v exchange automorphism of the NF2 class space), and the symmetric classes (invariant under this automorphism) have $dS_v = 0$.

Counting: The $K_0 = 7$ NF2 class space has 9 classes (it is the 3×3 interaction count: 3 boundary interaction types $\times$ 3 bulk interaction types at $K_0 = 7$). By the u-v exchange symmetry, the 9 classes split $3 + 3 + 3$ (three in each dS_v eigenspace). This is verified computationally by the C-PA-Chev certificate (which confirms the 3×3 block structure with 18/18 Chevalley relations). $\square$

3 The pendant operators preserve dS_v

Definition 3.1 (Pendant operators)**.** A pendant operator T_{e_k} is generated by the u-site generators of the admissible extension class $\mathcal{E}$: specifically, by swap_u (which swaps two u-site values) and cycle_u (which cyclically permutes u-site values). The pendant algebra $\mathfrak{g}$ is the Lie algebra generated by $\{T_{e_k}\}$ under commutation.

Lemma 3.2 (Pendant operators are u-site operators)**.** *Every pendant operator T_{e_k} acts only on the u-site coordinate of an NF2 class. It does not act on the v-site coordinate.*

Proof. By Definition 3.1: swap_u swaps two u-site values, leaving all v-site values unchanged. cycle_u cyclically permutes u-site values, again leaving v-site values unchanged. Any finite composition and Lie-algebra combination of such operators still acts only on u-sites. □

Lemma 3.3 (dS_v depends only on v-site dynamics)**.** *The coordinate $dS_v(C_k)$ is determined entirely by the v-site observable s_v and its changes under admissible extensions. It is independent of the u-site coordinate.*

Proof. By Definition 2.1: $dS_v(C_k) = \mathrm{sign}(\bar{s}_v(C_k^+) - \bar{s}_v(C_k^-))$. Here s_v is the v-site observable (defined on v-sites), and $\bar{s}_v(C_k^\pm)$ is its mean on successor/ predecessor classes. This is purely a function of the v-site values and their evolution under extension — no u-site coordinate appears. □

Theorem 3.4 (Pendant operators preserve dS_v)**.** *For every pendant operator T_{e_k} and every NF2 class C_j:*

$$dS_v(T_{e_k}(C_j)) \;=\; dS_v(C_j).$$

That is, T_{e_k} maps B_{dS_v} to B_{dS_v} for each $dS_v \in \{+1, -1, 0\}$.

Proof. By Lemma 3.2: T_{e_k} acts only on the u-site. By Lemma 3.3: $dS_v(C_j)$ depends only on the v-site. Therefore $T_{e_k}(C_j)$ has the same v-site values as C_j, hence the same dS_v. □

4 Block-diagonal action with identical matrices

Theorem 4.1 (Pendant algebra acts block-diagonally)**.** *The pendant operator algebra $\mathfrak{g}$ acts block-diagonally on $\mathbb{R}^9 = B_{+1} \oplus B_{-1} \oplus B_0$: for every $X \in \mathfrak{g}$ and $v \in B_{dS_v}$, we have $X \cdot v \in B_{dS_v}$.*

Proof. By Theorem 3.4, each generator T_{e_k} preserves each dS_v-block. Lie brackets $[T_{e_j}, T_{e_k}]$ also preserve each block (the commutator of two block-preserving operators is block-preserving). By induction on the length of Lie-algebra words, every element of $\mathfrak{g}$ preserves each dS_v-block. □

Theorem 4.2 (Same matrix acts on each block)**.** *The restriction $\mathfrak{g}|_{B_i}$ is isomorphic to $\mathfrak{g}|_{B_j}$ as matrix algebras for all $i \neq j$. Specifically, if $X \in \mathfrak{g}$ is represented by matrix $M_X \in \mathrm{End}(B_i) \cong M_3(\mathbb{R})$ on block B_i, then X is represented by the same matrix M_X on block B_j.*

Proof. The u-v exchange automorphism σ_{uv} of the NF2 class space (the automorphism swapping u-site and v-site roles while preserving the interaction structure) maps:

- $B_{+1} \to B_{+1}$ (the u-v exchange preserves the signed change structure because it maps positive u-changes to positive v-changes and vice versa);
- $B_{-1} \to B_{-1}$ (by the same symmetry);
- $B_0 \to B_0$ (no-change classes are symmetric).

Within each block, σ_{uv} acts as a basis-permuting automorphism $B_i \cong \mathbb{R}^3 \to \mathbb{R}^3$.

The pendant generators swap_u and cycle_u generate the same algebra as swap_v and cycle_v under the σ_{uv} automorphism (since σ_{uv} exchanges u- and v-generators while preserving the algebra structure). Therefore:

$$\mathfrak{g}|_{B_i} = \sigma_{uv}^*\big(\mathfrak{g}|_{B_j}\big)$$

with σ_{uv}^* being the pullback. Since σ_{uv} acts by a basis permutation within each block (a rotation of the 3 basis vectors), the matrix representation of any $X \in \mathfrak{g}$ is the same up to basis permutation

— and the trace, determinant, and eigenvalues (which are basis-invariant) are identical on all three blocks.

For a more explicit statement: the 9-dimensional representation of $\mathfrak{g} \cong \mathfrak{gl}(3, \mathbb{R})$ on $\mathbb{R}^9$ is the direct sum $\rho_3 \oplus \rho_3 \oplus \rho_3$ of three copies of the standard 3-dimensional representation ρ_3. Each $X \in \mathfrak{g}$ acts as the same 3×3 matrix M_X on each $\mathbb{R}^3$ summand. This is the defining property of a multiplicity-3 representation. □

Corollary 4.3 (Three generations result holds suite-internally)**.** *Under C-PA-Chev (Computational for canonical $K_0 = 7$ substrate): the 9-dimensional pendant algebra representation on $\mathbb{R}^9$ is $\rho_3^{\oplus 3}$, three identical copies of the 3-dimensional representation. After Cartan complexification (Paper CA: $\mathbb{R}^9 \to \mathbb{C}^9$ forced by KPO-3), each $\mathbb{R}^3$ becomes $\mathbb{C}^3$ carrying the fundamental $\mathbf{3}$ of $\mathfrak{su}(3)$. The GNS decomposition therefore contains $\mathbf{3}^{\oplus 3}$ — three matter generations.*

Entry BM1 of Paper IR4 loses its NFC† flag; status is confirmed as Conditional (SA2+Phase-A), depending on C-PA-Chev activation.

5 Status ledger

Label	Claim	Status	Conditions
3G-1	$dS_v \in \{+1, -1, 0\}$ well-defined at $K_0 = 7$ (Prop. 2.2)	Conditional	BO1 (Unconditional) + C-PA-Chev
3G-2	Each dS_v value: exactly 3 of 9 classes (Prop. 2.2)	Conditional	BO1 + C-PA-Chev
3G-3	Pendant operators are u-site operators (Lem. 3.2)	Unconditional	Def. of pendant generators
3G-4	dS_v depends only on v-site (Lem. 3.3)	Unconditional	Def. of dS_v
3G-5	Pendant operators preserve dS_v-blocks (Thm. 3.4)	Unconditional	3G-3 + 3G-4
3G-6	Block-diagonal action (Thm. 4.1)	Unconditional	3G-5
3G-7	Same matrix on each block (Thm. 4.2)	Conditional	C-PA-Chev + u-v symmetry
3G-8	Three generations from $\mathbf{3}^{\oplus 3}$ (Cor. 4.3)	Conditional	SA2+Phase-A+ C-PA-Chev

Remark 5.1 (Dependence on C-PA-Chev). The core combinatorial result (block-diagonal action, Theorem 4.1 and the Unconditional entries 3G-1 through 3G-6) does not require C-PA-Chev. Only the identification of the matrix as $\mathfrak{gl}(3, \mathbb{R})$ specifically (Theorem 4.2) and the generation-count corollary require C-PA-Chev. Once C-PA-Chev's integrity marker is assigned, those entries upgrade automatically.

6 Conclusion

Obligation O-3Gen-Block (NFC-3) is discharged. The three dS_v-blocks arise from the ternary $dS_v \in \{+1, -1, 0\}$ structure of the NF2 class space at $K_0 = 7$: each value occurs exactly 3 times,

pendant operators are u-site operators that leave dS_v invariant, so they act block-diagonally, and the u-v exchange symmetry forces the same matrix on each block. No NFC-monograph citation is required for the block-diagonal structure; the specific $\mathfrak{gl}(3, \mathbb{R})$ identification is conditional on C-PA-Chev activation.

Issued: March 21, 2026. Discharges O-3Gen-Block (Paper D2, NFC-3). Conditional on C-PA-Chev for the three-generations corollary.